

Real-time analysis of foodborne outbreaks: Bongkreikic acid poisoning in Taiwan, 2024

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Introduction

In March 2024, Taiwan experienced its first documented Bongkreikic acid (BA) poisoning outbreak, affecting 33 individuals who dined at a restaurant in Taipei. Early outbreak detection and accurate prediction of its potential scope are crucial for effective public health responses. While nowcasting methods have been successfully applied to infectious disease outbreaks, their application to foodborne illness outbreaks presents unique challenges that warrant investigation.

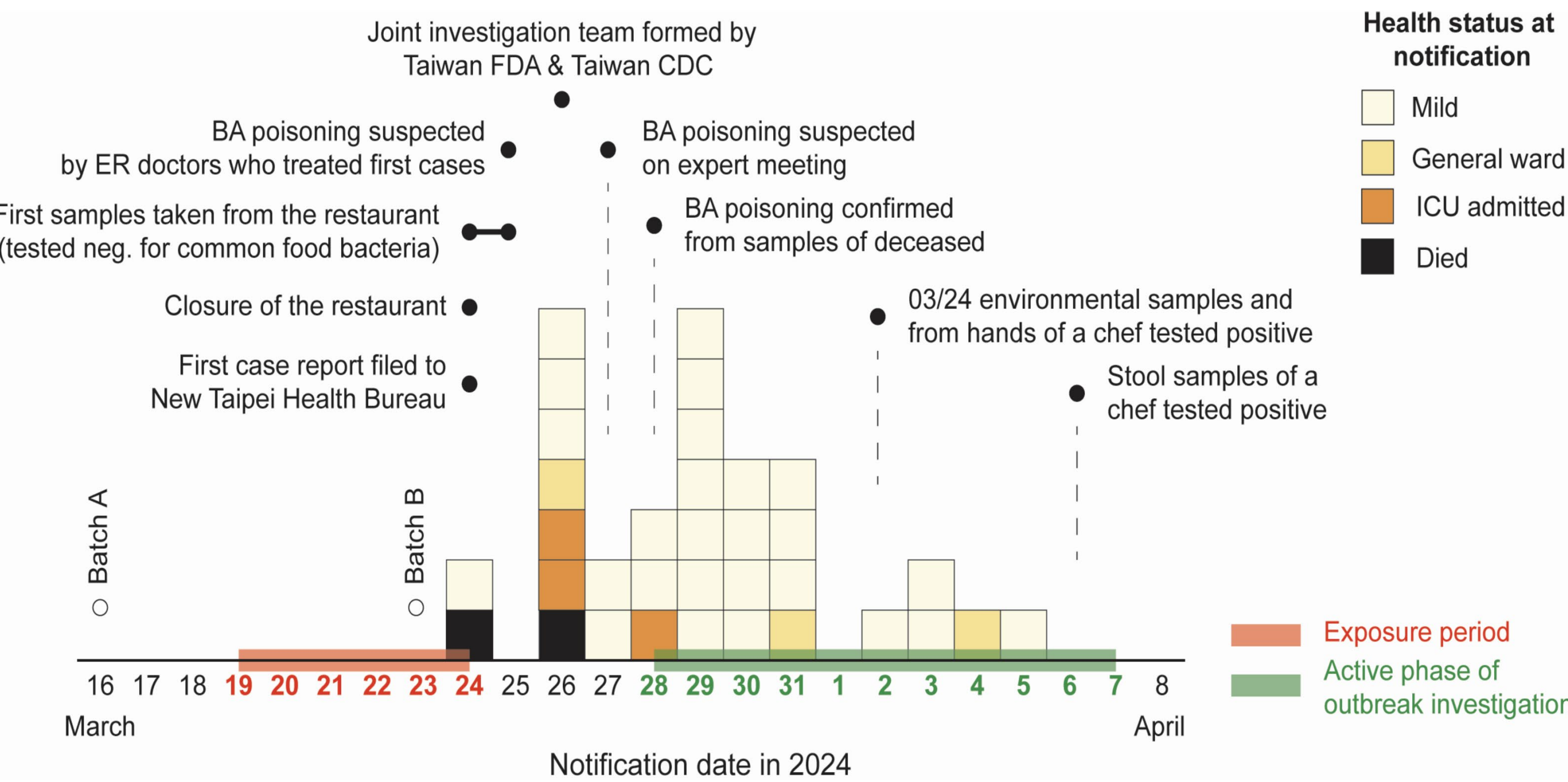


Figure 1: Timeline of Bunkerik Acid (BA) Poisoning Outbreaks in Taiwan from March to April 2024

Materials & Methods

Data were compiled from regular reports by the Department of Health of Taipei City Government and Taiwan Centers for Disease Control.

The statistical analysis was conducted using three approaches. The time intervals from infection to healthcare seeking and case reporting were modelled using Generalized Gamma distributions (GGD), with means and standard deviations estimated through model fitting. Secondly, attack rates (AR) were calculated using a binomial model to quantify the probability of illness after dining on specific dates. Thirdly, case-fatality ratios (CFR) were estimated using binomial distributions and compared with historical outbreak data from Indonesia and mainland China.

Results

The initial AR calculations showed significant daily variation (43.8-80.4%), with an overall average of 60.3%.

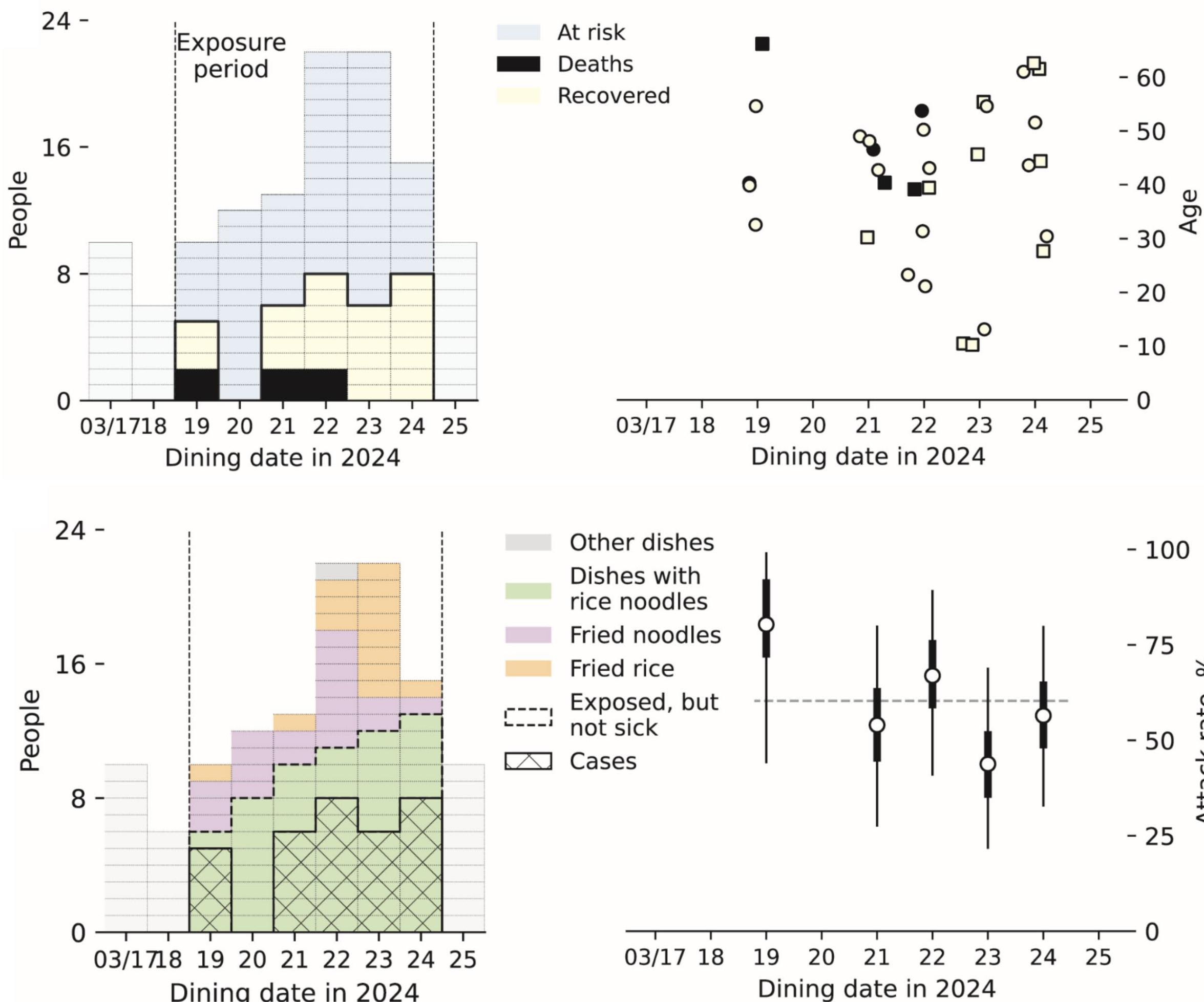


Figure 2: Top Left: Displays the daily number of cases and at-risk individuals during the exposure period (March 19th-24th, 2024).Bottom Left: Shows the meal preferences of restaurant customers, with cases marked in shaded boxes. Bottom Right: Circles represent females, and squares represent males. Bottom Right: Daily attack rate (AR) estimates, with circles indicating post-test averages, thin lines indicating 95% confidence intervals, and thick lines indicating post-test interquartile ranges. The dotted line indicates the average AR for all five days (excluding 20 March).

The case-fatality ratio reached 20.0% (95% CI: 8.7%-34.7%), with particularly severe outcomes among intensive care unit patients where six of seven admitted patients died. This CFR falls between historical records from other regions, being higher than Indonesia's documented rate of 14.0% but lower than mainland China's 39.5%.

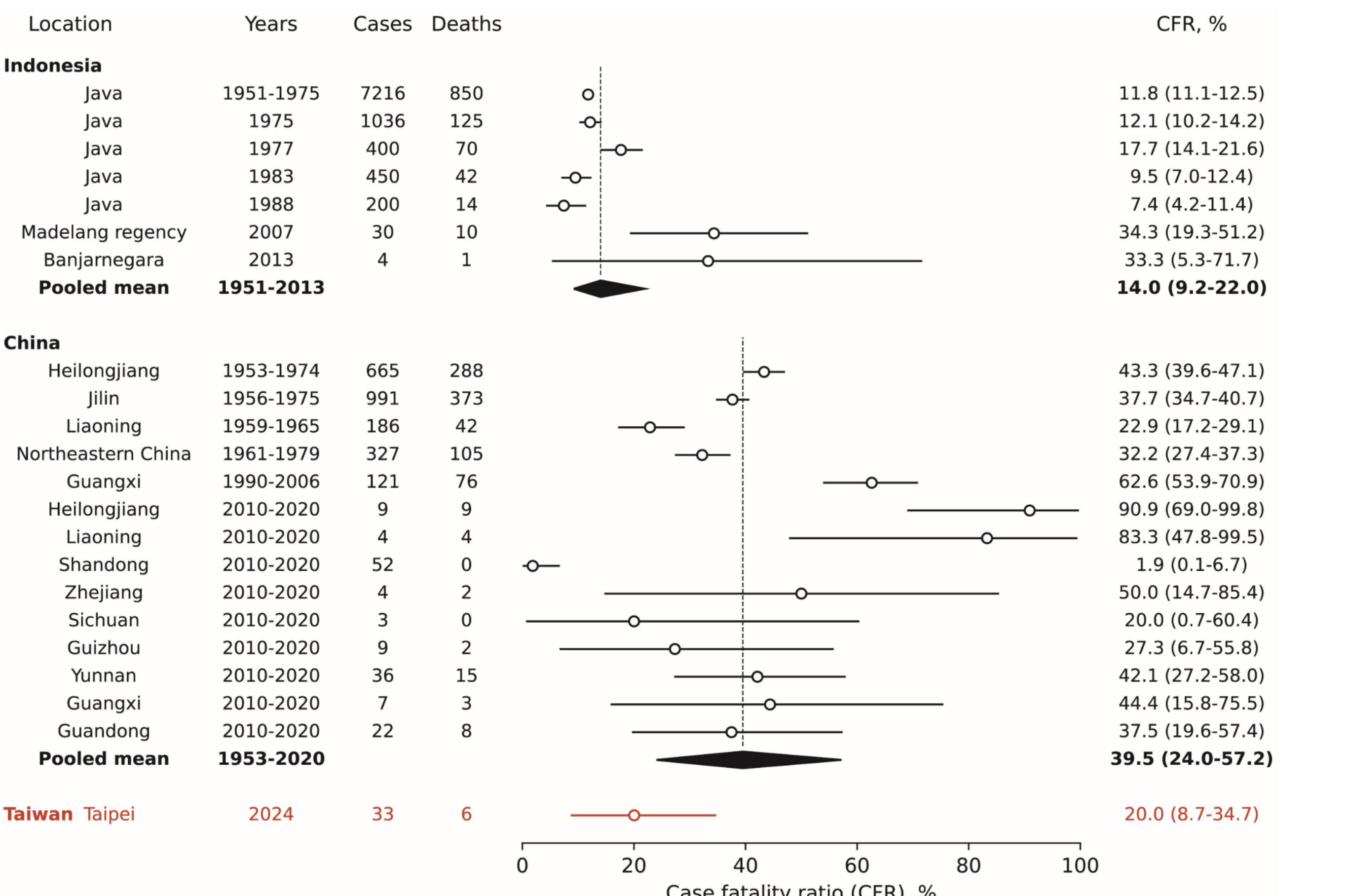


Figure 3: Historical estimates of the case fatality rate (CFR) of BA poisoning outbreaks.

Discussion

In this outbreak, the primary source of food-borne illness has been identified as flat rice noodle. When commercially manufactured products are involved, contamination can occur at either the manufacturing facility or during food preparation, which significantly complicates source identification.

Figure 2 shows that although many customers were potentially at risk on the 20th, no cases were reported on that day. This suggests that either the noodles came from a different, uncontaminated batch, or that food handling procedures (cooking time, temperature, storage) were different. This complexity necessitates a comprehensive approach to early warning systems that can account for multiple potential contamination points along the supply chain.

Unlike infectious diseases where R_0 values help predict transmission patterns, foodborne outbreaks follow point-source transmission patterns and are complicated by time lags between exposure, symptom onset, and case confirmation.

Therefore, the application of nowcasting to these small-scale outbreaks presents significant challenges due to the complex interplay of multiple variables in the food supply chain, the difficulty in pinpointing contamination sources, and the potential for rapid geographical spread of affected products, making accurate prediction and early intervention particularly challenging in these scenarios.

Acknowledgements

This work was conducted under the supervision of Dr. Andrei R. Akhmetzhanov in collaboration with Zhen-Zhi Chu, Bianca de Padua, and Dr. Robin N. Thompson.

Funding

This work was supported by the National Science and Technology Council (grant #113-2314-B-002 -181 -MY3)

Reference

Akhmetzhanov AR, de Padua B, CS Wong, Thompson RN. First documented outbreak of Bongkreikic acid food poisoning in Taiwan, March-April 2024. In preparation.